

The cross-base identity for OEIS A126409, proved

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Abstract

OEIS A126409 carries an empirical identity, in two clauses, relating its counts to those of the entry one base smaller, contributed by R. H. Hardin. Both clauses are true and both are proved here, with no computation in the argument. The difference between consecutive bases counts the admissible strings that USE the new largest digit; those are the translates of the translation-classes of admissible strings over $\mathbf{Z}$ whose range fits below the top of the alphabet; the classes are the step vectors, of which there are $(2d + 1)^{n-1}$; and at the one base where the fit fails by exactly one, the classes lost are the two monotone ones, which is the entry's own -2 .

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1 The conjecture

OEIS A126409 is “Number of base 22 n -digit numbers with adjacent digits differing by two or less.”. It has offset 0 and begins

1, 22, 104, 500, 2418, 11736, 57098, 278274, ...

The entry states, as an empirical claim and never marked settled:

[Empirical] $a(\text{base}, n) = a(\text{base}-1, n) + 5^{n-1}$ for $\text{base} \geq 2n-1$; $a(\text{base}, n) = a(\text{base}-1, n) + 5^{n-1} - 2$ when $\text{base} = 2n-2$

As of the “Last modified” line on the live entry (Aug 12 2026 19:11:51, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

For this entry the base is 22 and the permitted difference is $d = 2$, so the first clause claims $a_{22}(n) - a_{21}(n) = 5^{n-1}$ whenever $2(n-1) + 1 \leq 22$, that is for $1 \leq n \leq 11$, and the second claims $a_{22}(n) - a_{21}(n) = 5^{n-1} - 2$ at the single n with $2(n-1) = 22$.

2 What the entry counts

A base- b n -digit string in this family's sense is a tuple $(d_1, \dots, d_n) \in \{0, 1, \dots, b-1\}^n$ with $|d_i - d_{i+1}| \leq d$ for $1 \leq i < n$; there is no wrap-around. Leading zeros are counted, which is what the entry's $a(1) = b$ says, and $a(0) = 1$ is the empty string. Write $L_d(b, n)$ for their number. Evaluated exactly in integer arithmetic, L_2 at base 22 reproduces all 21 terms A126409 publishes.

3 The identity

Theorem 1. Fix $d \geq 1$ and $n \geq 1$, and put $K = 2d + 1$.

(i) If $b \geq d(n - 1) + 1$ then $L_d(b, n) - L_d(b - 1, n) = K^{n-1}$.

(ii) If $b = d(n - 1)$ then $L_d(b, n) - L_d(b - 1, n) = K^{n-1} - 2$.

Proof. Step 1. The admissible strings over $\{0, \dots, b - 1\}$ that do not use the value $b - 1$ are exactly the admissible strings over $\{0, \dots, b - 2\}$, so $L_d(b, n) - L_d(b - 1, n)$ is the number of admissible strings over $\{0, \dots, b - 1\}$ in which $b - 1$ occurs.

Step 2. Consider the admissible strings over $\mathbf{Z}$: tuples $(c_1, \dots, c_n) \in \mathbf{Z}^n$ with $|c_i - c_{i+1}| \leq d$. Translation by an integer acts freely on them, each orbit contains exactly one representative with $c_1 = 0$, and such a representative is determined by its step vector $(s_1, \dots, s_{n-1})$, $s_i = c_{i+1} - c_i$, which may be any element of $\{-d, \dots, d\}^{n-1}$. So there are exactly K^{n-1} orbits. Each orbit also contains exactly one representative with $\max_i c_i = 0$, and all its representatives share one RANGE $\rho = \max_i c_i - \min_i c_i$, which satisfies $\rho \leq d(n - 1)$ because each of the $n - 1$ steps moves by at most d .

Step 3. A string counted in Step 1 has maximum $b - 1$ and entries at least 0, so it is the representative with maximum $b - 1$ of an orbit of range at most $b - 1$; conversely every orbit of range at most $b - 1$ gives exactly one such string. Hence

$$L_d(b, n) - L_d(b - 1, n) = \#\{\text{orbits of range } \leq b - 1\}.$$

Step 4. If $b \geq d(n - 1) + 1$ then $b - 1 \geq d(n - 1)$ and by Step 2 every orbit qualifies, giving (i). If $b = d(n - 1)$ then exactly the orbits of range $d(n - 1)$ are excluded. An orbit has range $d(n - 1)$ only if some c_i attains the maximum, some c_j the minimum, and the walk between them covers $d(n - 1)$; that walk has at most $n - 1$ steps, each of size at most d , so it must use all $n - 1$ of them in the same direction at full size. There are exactly two such step vectors, $(d, \dots, d)$ and $(-d, \dots, -d)$, so two orbits are lost and (ii) follows. $\square$

4 Verification

Three checks, each independent of the proof above.

First, the model reproduces all 21 terms A126409 publishes, exactly, in integer arithmetic; the reading of the entry's English is what is being tested there.

Second, the first clause was evaluated directly for every n in the entry's claimed range, $1 \leq n \leq 11$, both sides computed from the definition and not from the theorem:

n	$L_2(22, n)$	$L_2(21, n)$	5^{n-1}
1	22	21	1
2	104	99	5
3	500	475	25
4	2418	2293	125
5	11736	11111	625
6	57098	53973	3125
7	278274	262649	15625
8	1357980	1279855	78125

and the difference of the second and third columns is the fourth at every row shown and at every row up to $n = 11$.

Third, the second clause was evaluated at its own index. For this entry the second clause bites at $n = 12$, where $b = 22 = 2(n - 1)$: there $L_2(22, 12) - L_2(21, 12) = 48828123$ and $5^{11} - 2 = 48828123$.

References

1. The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A126409>.